

Supplementary Information

Delta Dependency Prioritizes Paralog-Based Synthetic Lethality Candidates Across Solid Tumor Types

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1 Supplementary Methods

1.1 Gold-standard curation and tier assignment

Each of the twelve curated pairs was verified against its primary citation(s) before inclusion. For every pair we recorded the assay type, model system, validated direction, whether the evidence derives from simultaneous dual-gene perturbation, whether it is independent of DepMap CRISPR data, and whether it directly demonstrates synthetic lethality in the direction scored here (Table S3). Pairs were then assigned to three evidence tiers. Tier A requires direct genetic synthetic-lethal evidence from dual-gene perturbation (combinatorial or digenic CRISPR knockout; 3 pairs). Tier B requires that the paralog be a demonstrated selective dependency in driver-mutant cells from single-gene perturbation with functional validation (natural-genotype conditional dependency; 2 pairs). The Tier A $\cup$ Tier B set constitutes the pre-specified primary external benchmark. Tier C comprises indirect evidence only — reciprocal-direction-only evidence (EP300 $\rightarrow$ CREBBP), evidence supporting a different driver (PIK3CA $\rightarrow$ PIK3CB, whose primary evidence supports PTEN $\rightarrow$ PIK3CB), developmental redundancy (CCNE1 $\rightarrow$ CCNE2), or DepMap computational analyses (FBXW7 $\rightarrow$ FBXW2, PPP2R1A $\rightarrow$ PPP2R1B) — and is excluded from the primary benchmark (5 pairs). Two further pairs that are not sequence paralogs (BRCA1 $\leftrightarrow$ BRCA2, functional analog; STK11 $\rightarrow$ SIK1, partial homolog/pathway axis) serve as mechanistic comparators and specificity references only. Pharmacologic inhibition of both members of a gene family was not accepted as sufficient evidence of genotype-conditional synthetic lethality in a specified direction.

1.2 Driver-mutation classification

Driver-mutation status was defined per gene class on the default omics profile of each DepMap 26Q1 model (`IsDefaultEntryForModel1`). Tumor suppressors (TSG) require a DepMap-curated likely loss-of-function variant (`LikelyLoF`; includes dominant-negative TP53 missense); oncogenes require a recurrent oncogenic hotspot variant (`Hotspot`). Gene classes were assigned by the DepMap 26Q1 `OncogeneHighImpact`/`TumorSuppressorHighImpact` majority vote, with one documented manual override (GATA3 as TSG). Amplification-driven oncogenes without qualifying hotspot mutations (e.g., CCNE1) contribute no mutant lines under this rule and are analysed through copy-number channels instead. The per-gene comparison of the permissive non-silent definition versus the class-specific rule, with retained cell line counts and the top `VariantInfo` terms, is given in Table S6 (machine-readable mirror: `TableS6_MutationRules.tsv`).

1.3 DD computation and evaluation frames

DD was computed per driver $\times$ paralog $\times$ lineage stratum as the difference in mean Chronos gene-effect scores between driver-mutant and wild-type cell lines (Equation 1 of the main text), with Welch's *t*-test (unequal variances); effect sizes are reported as Cohen's *d* (pooled SD) and Hedges' *g* (small-sample corrected; Table S7). The primary analysis frame requires ≥ 5 mutant and ≥ 5 wild-type cell lines per stratum (110 evaluable driver $\times$ paralog $\times$ lineage associations across the 12 curated pairs and matched controls; 17 columns in `TableS2_FullResults.tsv`); a relaxed $\geq 3/\geq 3$ frame was re-run as a sensitivity analysis (12 evaluable lineages versus 8 on the primary frame). Benjamini-Hochberg correction ($q < 0.20$) was applied within each driver gene across all of that driver's HGNC paralogs and lineages jointly; a stricter $q < 0.10$ cut was examined as sensitivity. This within-driver correction

is a discovery-stage device and does not constitute genome-wide FDR control.

1.4 Classifier training and cross-validation

For the head-to-head comparison on the 72-pair frame (6 known positives), four standard classifiers — logistic regression (LR; `max_iter=10,000`), random forest (RF; 500 trees, seed 42), SVM with RBF kernel, and SVM with linear kernel — were trained on five features ($|\text{DD}|$, PCS, $\Delta\text{Expression}$, necessity, mutation frequency), all with default scikit-learn 1.3.0 hyperparameters and no tuning. The composite score was excluded from classifier inputs because it is a deterministic function of the other features. Features were z-score standardized; evaluation used leave-one-pair-out cross-validation, scoring each held-out pair with the out-of-fold decision function (or predicted probability), and AUROC was computed from the pooled out-of-fold scores with the rank-based Mann–Whitney statistic. Standardized LR coefficients with Wald p -values were obtained from a full-data maximum-likelihood fit (statsmodels 0.14). With only 6 positive pairs these classifier results are noise-dominated and are reported as descriptive context, not as model selection.

1.5 CPTAC cohorts, missingness, and multiplicity

Seven CPTAC cohorts were analysed (BRCA $n = 122$, COAD $n = 97$, LUAD $n = 110$, GBM $n = 99$, PDAC $n = 140$, UCEC $n = 95$, LUSC $n = 108$; 771 tumours in total). Pearson correlations of $\log_2$ protein abundance were computed per paralog pair within each cohort, requiring ≥ 10 paired samples per test; samples lacking either protein measurement were excluded pairwise (no imputation). Significance used Benjamini–Hochberg correction within each pair across the 7 cohorts. The mutation-conditioned UCEC analysis compared ARID1B protein abundance between ARID1A-mutant ($n = 37$) and wild-type ($n = 58$) samples with the Wilcoxon rank-sum test. The ovarian HGSC global proteome (PDC study PDC000360, $n = 232$) was downloaded for reference but not used in the main analysis.

1.6 PRISM analysis

The PRISM Repurposing secondary screen (1,482 compounds $\times$ 727 cell lines) was analysed as driver-genotype-conditioned drug sensitivity: $\Delta\text{AUC} = \text{mean}(\log_2 \text{AUC in driver-mutant lines}) - \text{mean}(\log_2 \text{AUC in wild-type lines})$. Selectivity calls required BH $q < 0.25$, $\Delta\text{AUC} < 0$, and enrichment > 0.1 ; this deliberately permissive discovery-stage threshold admits up to $\sim 25\%$ expected false positives among passing associations. Drug-genotype associations do not imply direct binding of the paralog protein.

1.7 Dependency window score (DWS)

The DWS (Equation 2 of the main text) is the ratio of $|\text{DD}|$ to a conservative baseline-essentiality denominator, $\max(|\mu_{\text{Chronos}}|, f_{\text{pan-essential}}, 0.01)$, where $f_{\text{pan-essential}}$ is the fraction of lines with paralog Chronos score < -0.5 and the 0.01 floor prevents division by near-zero values. Selectivity is the mutant-minus-wild-type difference in the fraction of lines dependent on the paralog. Pairs were classified as PAN_ESSENTIAL ($f_{\text{pan-essential}} > 0.5$), HIGH_SELECTIVITY (selectivity > 0.15 and DWS > 1.0), MODERATE, or LOW_SELECTIVITY across 5 cancer contexts (Ovarian, Endometrial, Breast, Colorectal, PanCancer) on the ≥ 5 -mutant frame. Robustness was assessed by re-deriving rankings under alternative denominators ($|\mu_{\text{Chronos}}|$ only; $f_{\text{pan-essential}}$ only; their mean), alternative

floors (0.001, 0.05), and alternative selectivity thresholds (0.10, 0.20), and by bootstrap 95% confidence intervals for mean DWS and selectivity from 1,000 stratified resamples of cell lines within each context (seed 42; Tables S5 and S10).

1.8 Sequence- and structure-derived descriptors

Sequence features (length, molecular weight, pI, GRAVY hydrophobicity, disorder propensity, lysine/cysteine density) were computed from UniProt canonical sequences and domain architectures from UniProt annotations. For 13 of the 15 scored proteins, single-chain structures of the N-terminal 400 residues were predicted de novo with ESMFold v1 (mean pLDDT = 0.724); ARID1A and ARID1B ($\sim 2,200$ residues each) were assessed with sequence-level features only because the truncation covers only $\sim 17\%$ of the chain. Three heuristics are emphasized as exploratory: the feature-space structural similarity $\max(0, 1 - |\Delta_{\text{feature}}|)$ over the sequence features (not a 3D alignment metric); a sliding-window pocket count (windows with pLDDT > 0.7 and hydrophobic fraction 0.2–0.7, aggregated per 12 windows); and a PROTAC-suitability score (lysine density $\times 2.0$ + disorder propensity – cysteine density $\times 0.5$). The per-pair targetability score combined structural similarity (0.25), domain-architecture conservation (0.30), the druggability proxy (0.25), and the PROTAC score (0.20); the composite prioritization score combined max-normalized DWS (0.40), selectivity rescaled to $[0, 1]$ as $(s + 1)/2$ (0.30), and targetability (0.30), with all weights fixed a priori in `alphafold_analysis.py` and no calibration against drug-development data. Sequence identity was approximated by k -mer Jaccard similarity ($k = 3$), validated against Needleman–Wunsch global alignment identity on 50 protein pairs (Pearson $r = 0.88$; Fig. S9); the $\geq 30\%$ proxy threshold is an enrichment filter, not a precise identity cutoff.

2 Supplementary Figures

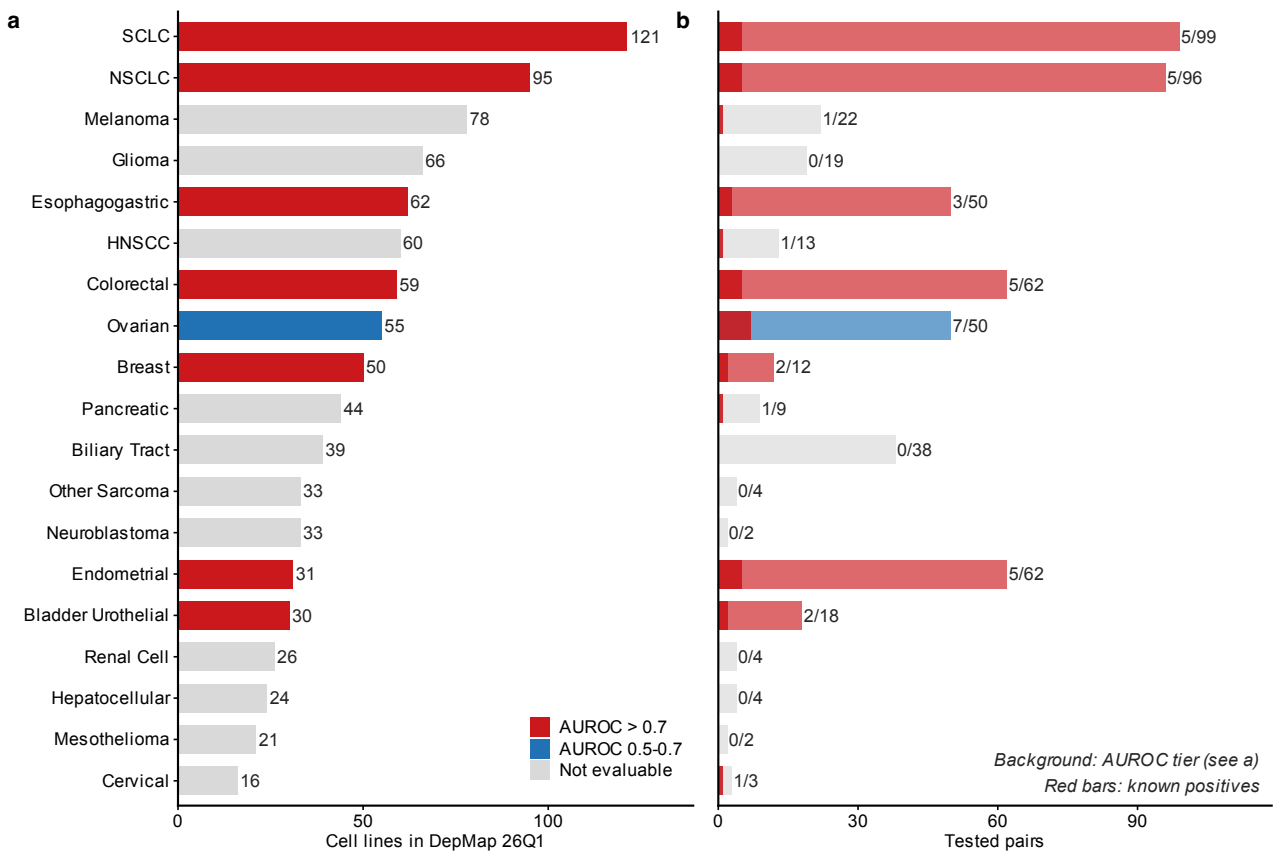


Fig. S1: Cell line landscape and evaluation summary across 23 solid tumor types. a, Cell line counts per cancer type, colored by DD AUROC tier (red: AUROC > 0.7; blue: 0.5–0.7; gray: < 0.5; light gray: not evaluable). b, Tested paralog pairs per cancer type; red bars show known gold-standard positives. Numbers indicate known/total pairs.

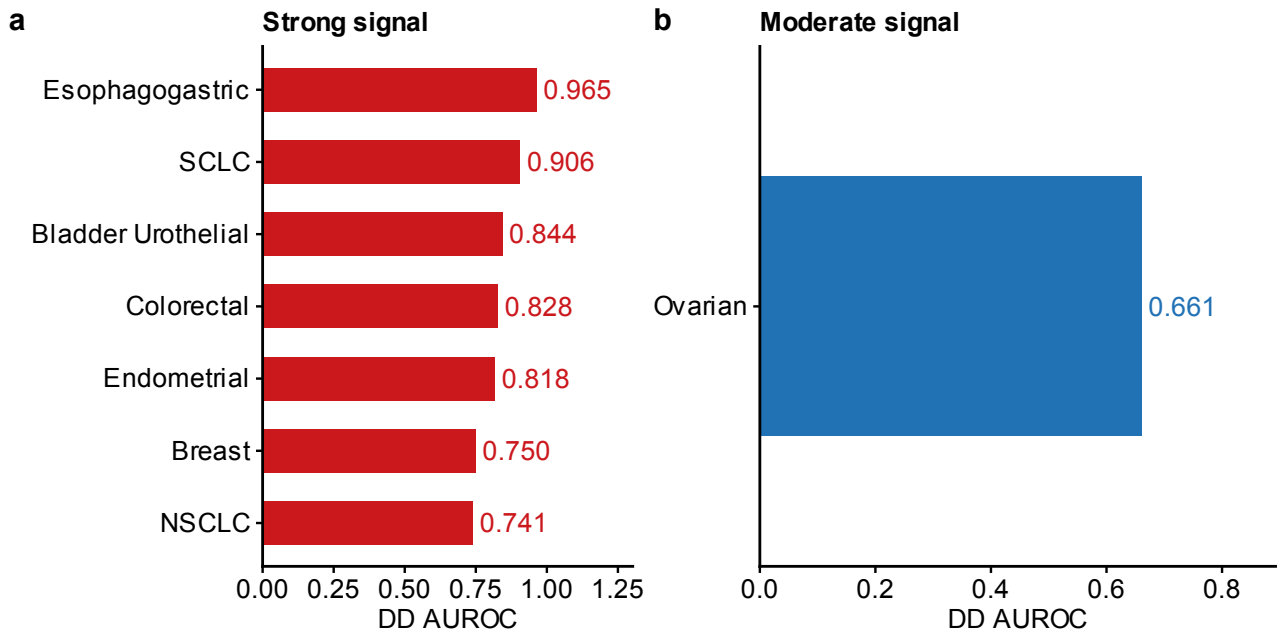


Fig. S2: Cross-cancer DD AUROC. DD AUROC for each of the 8 evaluable solid tumor types on the primary ≥ 5 -per-group frame, grouped by signal strength. a, Strong signal (AUROC ≥ 0.7 ; 7 lineages: Esophagogastric 0.965, SCLC 0.906, Bladder Urothelial 0.844, Colorectal 0.828, Endometrial 0.818, Breast 0.750, NSCLC 0.741). b, Moderate signal (AUROC 0.5–0.7; Ovarian 0.661). Dashed line indicates random classifier (0.5). Bar labels show the number of tested paralog pairs per lineage. Cancer types with < 2 known positive pairs were excluded from AUROC evaluation (see Table S1 for the 15 non-evaluable types). On the ≥ 3 -per-group sensitivity frame, four additional lineages become evaluable (Biliary Tract 0.990, Pancreatic 0.949, Melanoma 0.617, Cervical 0.500; 9 of 12 evaluable lineages exceed 0.7).

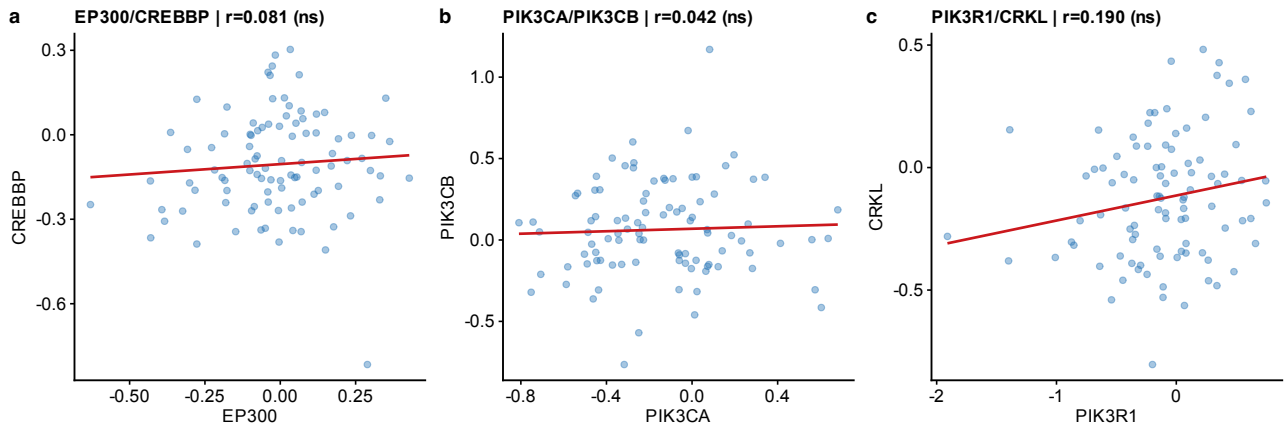


Fig. S3: CPTAC per-cohort scatter plots (UCEC). Driver vs. paralog protein abundance ($\log_2$ normalized intensity) in the UCEC cohort ($n = 83$): a, EP300 vs. CREBBP; b, PIK3CA vs. PIK3CB; c, PIK3R1 vs. CRKL. Each point represents one tumor sample. Line: linear regression fit with 95% confidence band. Pearson r is shown in each panel (ns: BH-corrected $q \geq 0.05$). Similar scatter plots for other cohorts and paralog pairs are available upon request.

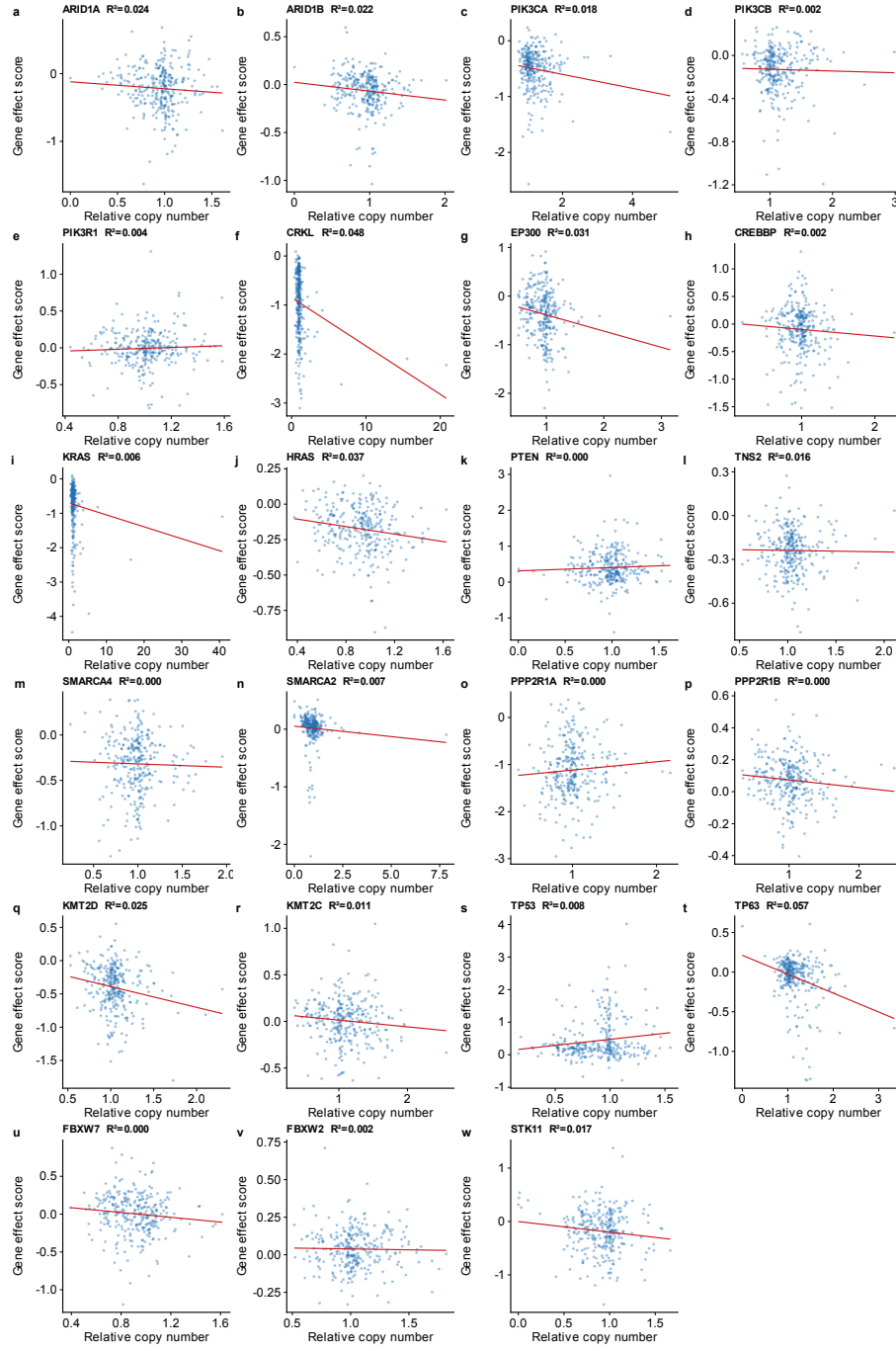


Fig. S4: CNV independence analysis. Scatter plots of relative copy number (x-axis) vs. Chronos gene-effect score (y-axis) for each of 23 paralog genes across 1,192 cell lines with both CNV and dependency data. Each panel shows one gene with the linear regression fit (red line) and R^2 value. All $R^2 < 0.10$, indicating that copy number variation explains $<10\%$ of gene-effect variance. Multivariate regression controlling for driver mutation and TP53 co-mutation left DD estimates essentially unchanged after CNV adjustment (see main text).

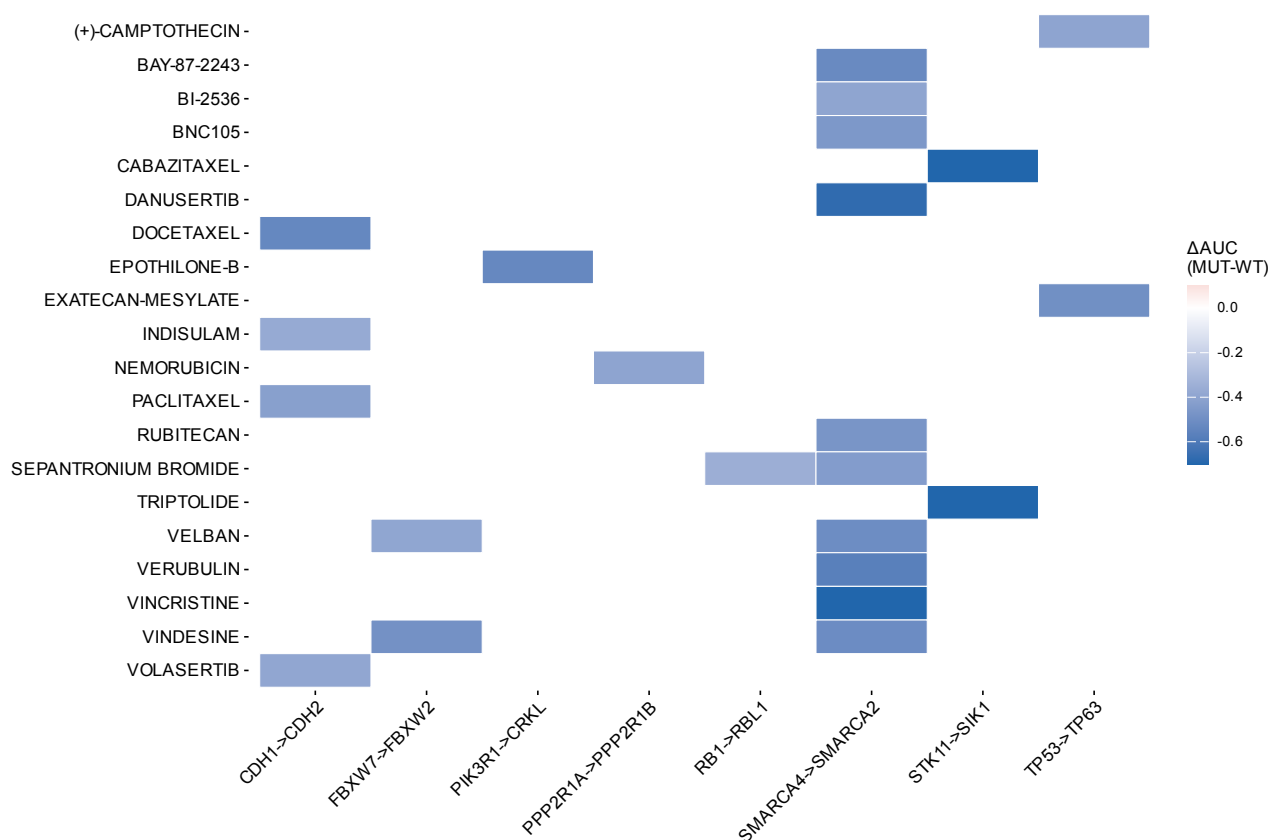


Fig. S5: PRISM drug selectivity heatmap. Heatmap of drug sensitivity differential (ΔAUC = mean AUC in driver-mutant minus mean AUC in wild-type cell lines) for the top driver-genotype-drug sensitivity associations from the PRISM Repurposing screen (1,482 compounds, 727 cell lines). Negative ΔAUC indicates selective sensitivity in driver-mutant cells. Associations shown met discovery-stage criteria (BH $q < 0.25$, $\Delta AUC < 0$, enrichment > 0.1). Known drug-target biology is recovered: MEK inhibitors with KRAS-mutant lines, mTOR/AKT inhibitors with PTEN-mutant lines, HDAC inhibitors with EP300-mutant lines. Drugs shown are not necessarily direct binders of the paralog protein.

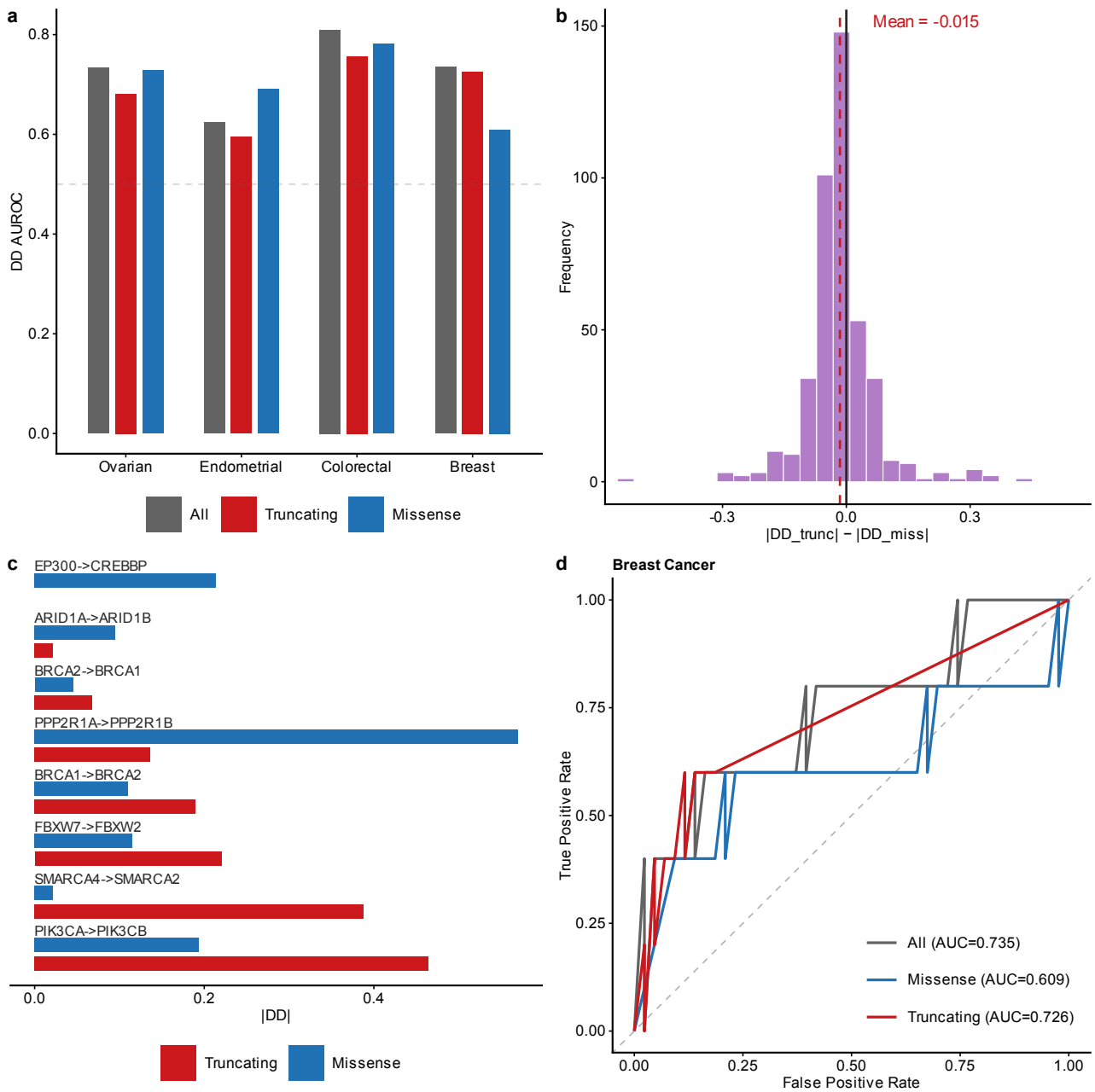


Fig. S6: Mutation-type stratification. a, DD AUROC stratified by mutation type (truncating vs. missense) across cancer types. b, Distribution of ΔDD (truncating minus missense) across all tested driver-paralog pairs; overall mean near zero (-0.015), indicating that truncating $>$ missense is pair-specific rather than universal. c, $|DD|$ under truncating (red) versus missense (blue) mutations for the known gold-standard SL pairs (maximum across cancer types, one row per pair); the truncating-missense gap is largest for ARID1A→ARID1B ($+0.368$ in Ovarian) and EP300→CREBBP ($+0.314$ in Colorectal). d, ROC curves for Breast cancer comparing truncating-only (AUROC = 0.726) vs. missense-only (0.391) and overall DD performance. Panel d is computed on the mutation-type analysis frame (48 Breast pairs); the resulting “All” AUC (0.735) therefore differs from the cross-cancer frame value (0.750; Fig. 1a), which uses a different pair universe.

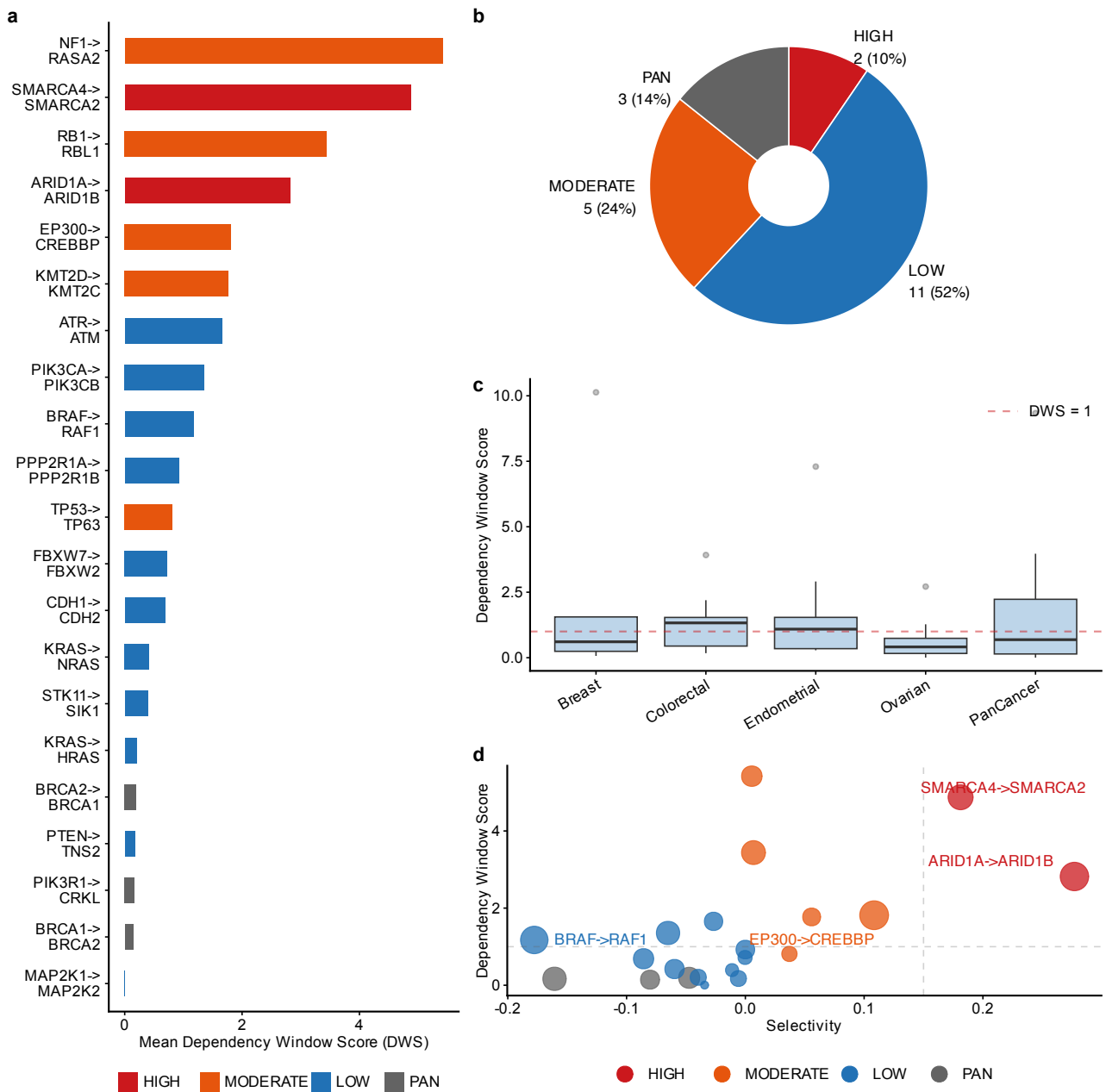


Fig. S7: Therapeutic window analysis. a, All 21 paralog pairs ranked by mean dependency window score (DWS) across up to 5 cancer contexts (Ovarian, Endometrial, Breast, Colorectal, PanCancer) on the ≥ 5 -mutant frame. SMARCA4→SMARCA2 (4.87) and ARID1A→ARID1B (2.82) are the two HIGH_SELECTIVITY pairs; NF1→RASA2 attains a higher raw DWS through a near-zero pan-essential denominator with selectivity ≈ 0 . b, In vitro selectivity tier classification: HIGH_SELECTIVITY (selectivity > 0.15 and DWS > 1.0 ; $n = 2$), MODERATE ($n = 5$), LOW_SELECTIVITY ($n = 11$), and PAN_ESSENTIAL ($f_{\text{pan-essential}} > 0.5$; $n = 3$). c, DWS values broken down by individual cancer context showing cross-lineage consistency for top candidates. d, Selectivity vs. DWS scatter plot; ARID1A→ARID1B and SMARCA4→SMARCA2 occupy the upper-right quadrant (high DWS and high selectivity).

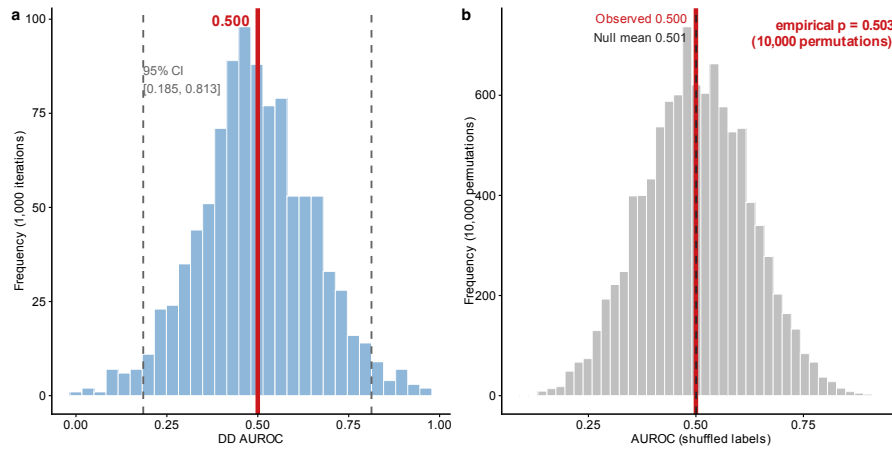


Fig. S8: Bootstrap and negative control (per-pair evaluation, 72 pairs, 6 known positives). a, Bootstrap distribution (1,000 iterations; 95% CI: 0.185–0.813). b, Null distribution from 10,000 label-shuffled permutations (empirical $p = 0.503$). The observed per-pair AUROC (0.500) does not exceed the null mean (0.501): with only 6 positive pairs the aggregated per-pair evaluation has no detectable signal, and the lineage-level evaluation (0.676, reported in the benchmark comparison) serves as the primary framework because it evaluates each driver $\times$ paralog $\times$ lineage combination separately rather than aggregating across lineages.

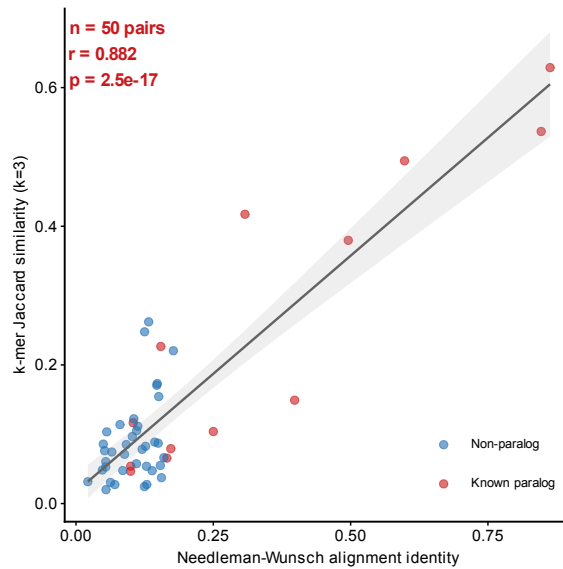


Fig. S9: k -mer Jaccard vs. Needleman-Wunsch alignment identity validation. Scatter plot of k -mer Jaccard similarity ($k = 3$) vs. global alignment identity for 50 protein pairs (13 known paralogs, 37 non-paralog controls). Red: known paralog-SL pairs; Blue: non-paralog pairs. Gray line: linear regression with 95% CI.

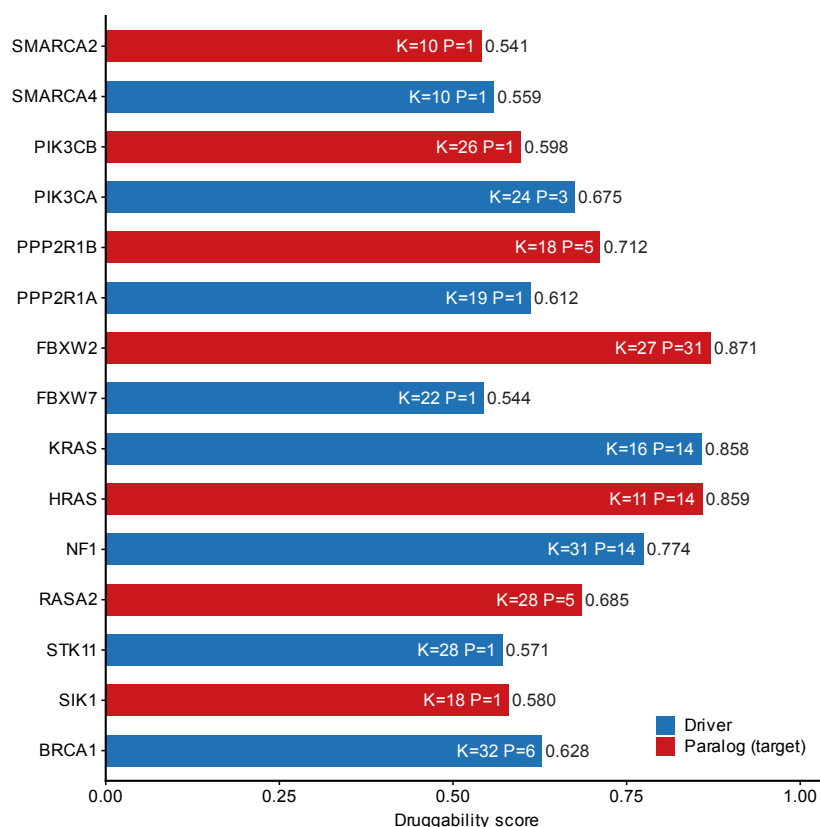


Fig. S10: Structure-based druggability assessment. Composite druggability score for 15 proteins (7 paralog targets in red, 8 driver genes in blue) based on ESMFold-predicted structures. Scores integrate structural confidence (pLDDT), hydrophobic content, lysine density (K, relevant for PROTAC conjugation), and predicted pocket count (P). For 13 of the 15 proteins, ESMFold v1 structures of the N-terminal 400 residues were predicted de novo (mean pLDDT = 0.724); ARID1A and ARID1B (~2,200 residues each) were assessed by sequence-level features only because the 400-residue truncation would cover only ~17% of the chain (Supplementary Table S9). FBXW2 (0.871) and KRAS/HRAS (0.858–0.859) show the highest scores due to well-folded compact domains.

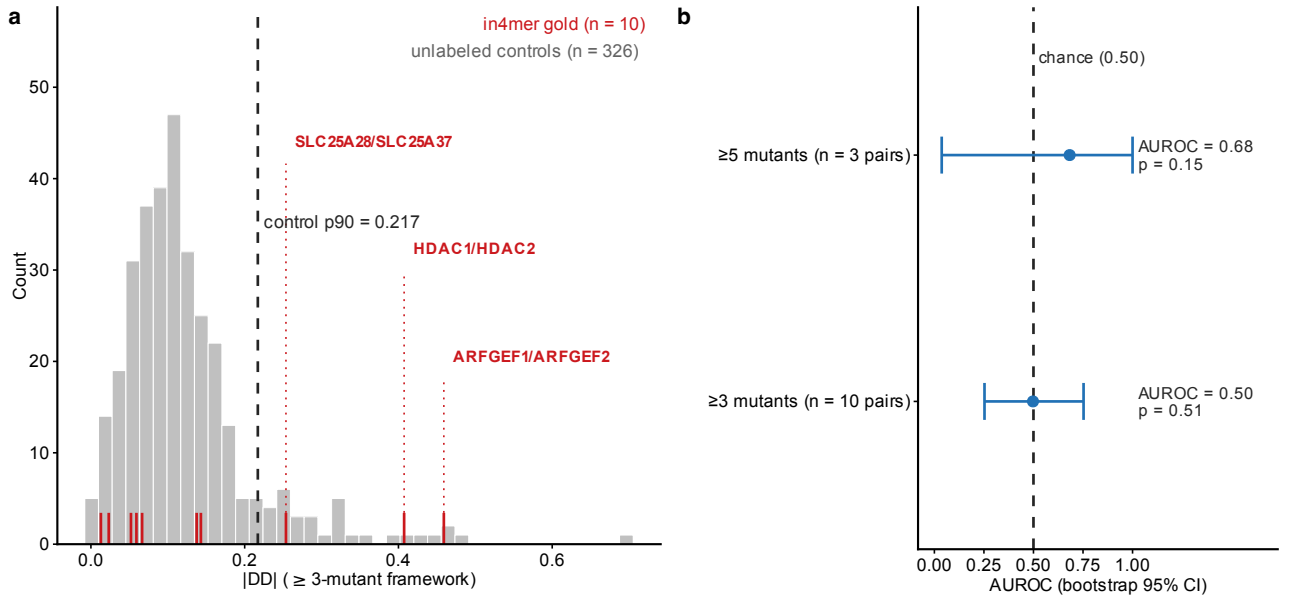


Fig. S11: External check against mutation-agnostic gold standards (in4mer). a, Distribution of $|DD|$ (≥ 3 -mutant framework) for 326 evaluable unlabeled control paralog pairs (gray histogram; 400 pairs sampled, seed 42). Red ticks: the 10 evaluable pairs of the 13 cross-platform in4mer gold standards (Esmaili Anvar et al., Nat Commun 2024). Dashed line: control 90th percentile (0.217); the three pairs above it (ARFGEF1/ARFGEF2, HDAC1/HDAC2, SLC25A28/SLC25A37) are those with the most recurrent natural mutations. b, AUROC point estimates with bootstrap 95% CI for both evaluation frameworks. Only 3 of the 13 pairs meet the ≥ 5 -mutant threshold (exploratory AUROC = 0.68); the 10 pairs evaluable at ≥ 3 do not separate from unlabeled controls (AUROC = 0.50, permutation $p = 0.51$), consistent with the mutation-agnostic, dual-knockout definition of these gold standards (main text, Limitations).

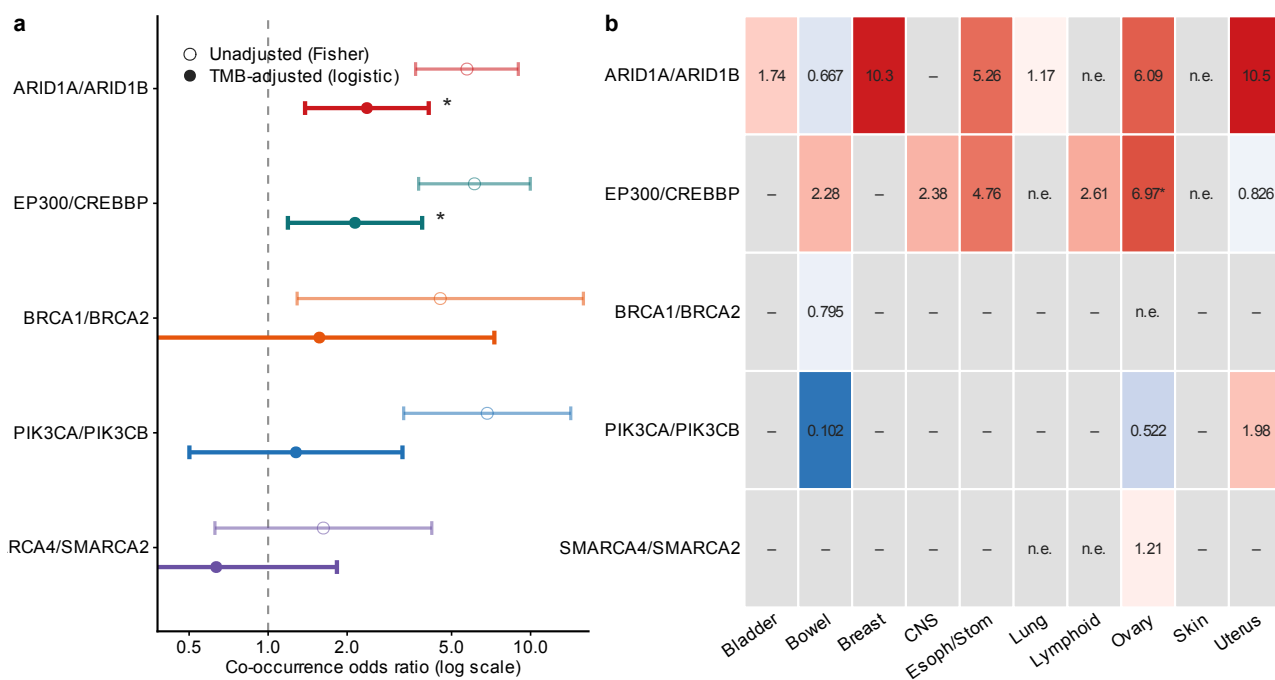


Fig. S12: Mutational co-occurrence of paralog pairs, with and without tumour mutational burden (TMB) adjustment. a, Pan-cancer co-occurrence odds ratios across 1,208 DepMap 26Q1 cell lines for the five testable pairs, unadjusted (Fisher’s exact test, open circles) vs. TMB-adjusted (filled circles; logistic regression of paralog mutation status on driver mutation status and $\log_e(1 + \text{TMB})$ per line). After TMB adjustment only ARID1A/ARID1B (adjusted OR = 2.38, $p = 0.002$) and EP300/CREBBP (adjusted OR = 2.14, $p = 0.011$) remain significant (*adjusted $p < 0.05$). b, Per-lineage TMB-adjusted odds ratios. “—”: not evaluable (< 30 lines or < 3 mutant lines per gene within the lineage); “n.e.”: not estimable (complete separation). Mutation status follows the gene-class-specific driver rules (Table S6). Source: `output/cooccurrence_by_lineage.json` (`cooccurrence_by_lineage.py`).

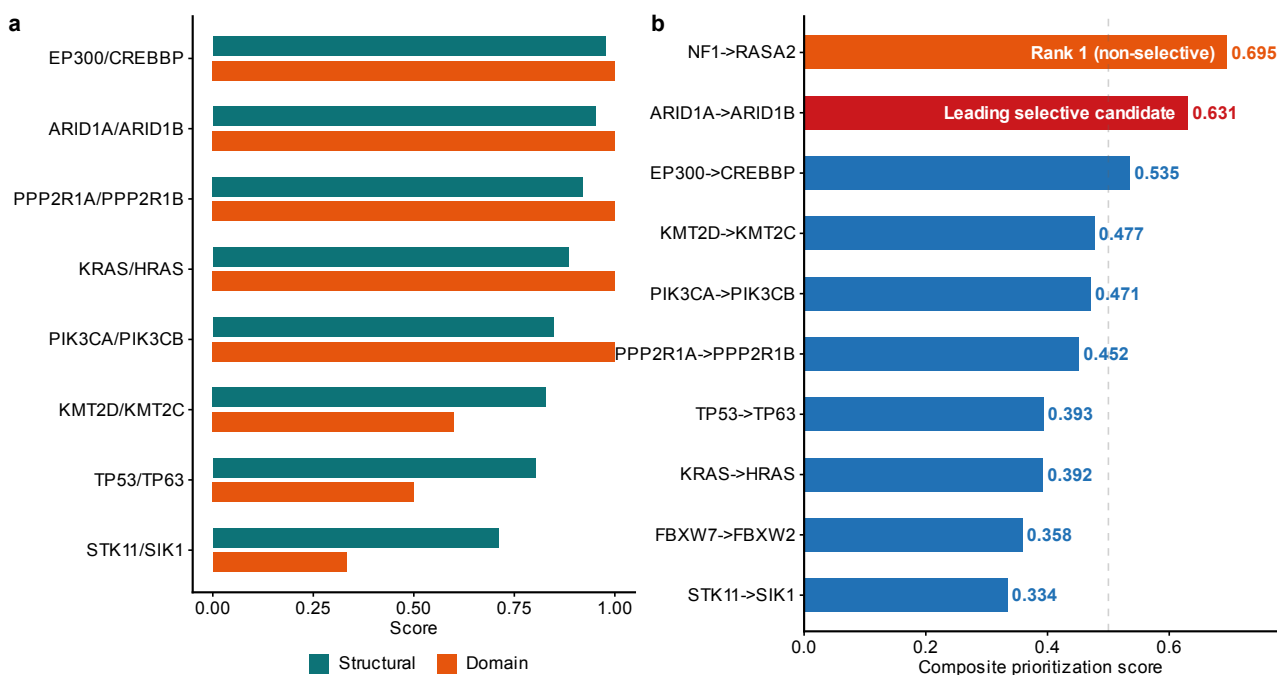


Fig. S13: Exploratory sequence- and structure-derived targetability descriptors. a, Feature-space structural similarity (Structural) and UniProt domain-architecture conservation (Domain) for the eight pairs with the highest structural similarity among the ten pairs with non-zero domain overlap (of 13 scored pairs); five of the 13 (38%) show perfect domain conservation. The structural similarity is a heuristic, $\max(0, 1 - |\Delta\text{feature}|)$ over sequence-level features, not a three-dimensional alignment metric (Methods). b, Composite prioritization score (max-normalized DWS 0.40, rescaled selectivity 0.30, targetability 0.30; weights fixed a priori in `alphafold_analysis.py`). NF1→RASA2 ranks first (0.695) only through a near-zero pan-essential denominator that inflates its raw DWS (selectivity ≈ 0); ARID1A→ARID1B (0.631) is the leading selective candidate. These descriptors are exploratory proxies and were not calibrated against drug-development data (main text, Methods and Limitations). Source: `output/alphafold_structural_analysis.csv` (R_figS13.R).

3 Supplementary Tables

3.1 Table S1: Cell line and evaluation summary across 23 solid tumor types

Label	Full name	Cell Lines	Pairs	Known+	DD AUROC
Esophagogastric	Esophagogastric adenocarcinoma / esophageal SCC	62	50	3	0.965
SCLC	Small cell lung cancer	121	99	5	0.906
Bladder Urothelial	Bladder urothelial carcinoma	30	18	2	0.844
Colorectal	Colorectal adenocarcinoma	59	62	5	0.828
Endometrial	Endometrial carcinoma	31	62	5	0.818
Breast	Invasive breast carcinoma	50	12	2	0.750
NSCLC	Non-small cell lung cancer	95	96	5	0.741
Ovarian	Ovarian epithelial carcinoma	55	50	7	0.661
Biliary Tract	Biliary tract cancer	39	38	0	—
Melanoma	Melanoma	78	22	1	—
Glioma	Diffuse glioma	66	19	0	—
HNSCC	Head and neck squamous cell carcinoma	60	13	1	—
Pancreatic	Pancreatic adenocarcinoma	44	9	1	—
Hepatocellular	Hepatocellular carcinoma	24	4	0	—
Renal Cell	Renal cell carcinoma	26	4	0	—
Other Sarcoma	Other soft-tissue sarcoma	33	4	0	—
Cervical	Cervical carcinoma	16	3	1	—
Mesothelioma	Pleural mesothelioma	21	2	0	—
Neuroblastoma	Neuroblastoma	33	2	0	—
Osteosarcoma	Osteosarcoma	19	0	0	—
Ewing Sarcoma	Ewing sarcoma	18	0	0	—
Rhabdomyosarcoma	Rhabdomyosarcoma	12	0	0	—
Thyroid	Thyroid carcinoma	10	0	0	—

Table S1: Cell line counts and DD AUROC across all 23 analyzed solid tumor types (primary ≥ 5 -per-group frame). Top: AUROC > 0.7 (7 types). Middle: AUROC ≤ 0.7 (1 type). Bottom: insufficient known positives for AUROC (15 types). Known+: number of gold-standard positive pairs evaluable in each lineage. “—”: AUROC not computable (< 2 known positives). Pairs: driver $\times$ paralog combinations passing the ≥ 5 mutant / ≥ 5 wild-type filter. On the relaxed ≥ 3 -per-group sensitivity frame, four additional lineages become evaluable (Biliary Tract 0.990, Pancreatic 0.949, Melanoma 0.617, Cervical 0.500), giving 12 evaluable lineages in total (9 above 0.7). Full name: standardized cancer-type name corresponding to each short label; SCC, squamous cell carcinoma.

3.2 Table S2: Complete paralog-SL pair analysis results

The full analysis results are provided as a tab-separated file: `TableS2_FullResults.tsv` (110 rows $\times$ 17 columns, representing all driver $\times$ paralog $\times$ cancer-type combinations passing the minimum sample-size filter of ≥ 5 mutant and ≥ 5 wild-type cell lines). Effect sizes (Cohen’s d , Hedges’ g) and per-stratum Welch t -test p -values for the same 110 associations are provided as `TableS7_EffectSizes.tsv` (Supplementary Table S7).

Data dictionary:

- **driver_gene:** Driver gene with damaging mutation.
- **paralog_gene:** Candidate paralog tested for conditional dependency.
- **cancer_type:** Solid tumor lineage (DepMap OncotreePrimaryDisease).
- **pcs:** Paralog Compensation Score (DD $\times$ necessity).
- **delta_expression:** Δ Expression between MUT and WT groups.
- **necessity:** Mean gene-effect score of the paralog (more negative = more essential).

- **dependency_dd**: Delta Dependency (signed; positive = greater dependency in MUT).
- **composite_score**: Weighted composite of PCS, DD, and Δ Expression.
- **novelty**: “Known” if in gold-standard set (Table S3), else “Novel”.
- **q_value**: Benjamini-Hochberg adjusted p -value within each driver gene.
- **mutation_frequency**: Fraction of cell lines with driver mutation in this lineage.
- **n_mut, n_wt**: Number of mutant and wild-type cell lines.
- **is_known_paralog_sl**: Boolean; TRUE if pair is in gold-standard set.

3.3 Table S3: Gold-standard paralog-SL pairs with evidence provenance and tier assignment

Tier	Driver	Paralog	Assay (perturbation)	Model system	Validated direction	di-	Dual?	Indep.	Direct SL	Inclusion	Key ref
A	AKT1	AKT2	Combinatorial CRISPR digenic KO	Cancer cell lines	AKT1→AKT2		Yes	Yes	Yes	Primary	Najm 2018
A	CDK4	CDK6	Digenic KO (pgPEN library)	Cancer cell lines	CDK4→CDK6		Yes	Yes	Yes	Primary	Parrish 2021
A	MAP2K1	MAP2K2	Digenic KO (pgPEN library)	Cancer cell lines	MAP2K1→MAP2K2		Yes	Yes	Yes	Primary	Parrish 2021
B	SMARCA4	SMARCA2	CRISPR KO conditioned on natural SMARCA4 mutation	SMARCA4-mutant cancer lines	SMARCA4→SMARCA2	No		Yes	Conditional	Primary	Hoffman 2014
B	ARID1A	ARID1B	shRNA knockdown conditioned on natural ARID1A mutation	ARID1A-mutant cancer lines	ARID1A→ARID1B		No	Yes	Conditional	Primary	Helming 2014
C	EP300	CREBBP	p300 degradation / CRISPR in CREBBP-deficient lines	CREBBP-mutant cancer lines	Reciprocal only (CREBBP→EP300)		No	Yes	Reciprocal	Secondary	Ogiwara 2016; Nie 2021
C	PIK3CA	PIK3CB	shRNA / PI3K inhibitor in PTEN-deficient lines	PTEN-deficient cancer lines	PTEN→PIK3CB		No	Yes	No (other driver)	Secondary	Wee 2008
C	CCNE1	CCNE2	Mouse developmental double knockout	Mouse embryo	CCNE1↔CCNE2	Yes (mouse)	Yes	Yes	No (redundancy)	Secondary	Geng 2003
C	FBXW7	FBXW2	DepMap computational analysis	700+ cell lines	FBXW7→FBXW2		No	No	No (computational)	Secondary	DepMap 26Q1 release
C	PPP2R1A	PPP2R1B	DepMap computational analysis	700+ cell lines	PPP2R1A→PPP2R1B		No	No	No (computational)	Secondary	DepMap 26Q1 release
comp.	BRCA1	BRCA2	PARP inhibition / genetic screen (functional analogs)	BRCA-mutant cancer lines	BRCA1/2→PARP axis		No	Yes	Functional analog	Comparator	Bryant 2005
comp.	STK11	SIK1	Mouse genetics, LKB1-SIK axis (partial homolog)	Mouse NSCLC models	STK11→SIK1/3 axis		No	Yes	Pathway axis	Comparator	Hollstein 2019

Table S3: Twelve curated pairs with evidence provenance and tier assignment after verification of the primary citations. **Tier A** (3 pairs): direct genetic synthetic-lethal evidence from dual-gene perturbation (combinatorial/digenic CRISPR knockout). **Tier B** (2 pairs): natural-genotype conditional dependency — the paralog is a demonstrated selective dependency in driver-mutant cells — from single-gene perturbation with functional validation. The Tier A $\cup$ Tier B set (5 pairs) constitutes the primary external benchmark. **Tier C** (5 pairs): indirect evidence only (reciprocal-direction-only, other-driver, developmental-redundancy, or DepMap-derived); excluded from the primary benchmark and reported separately. **Comparators** (2 pairs): mechanistic reference pairs that are not sequence paralogs (functional analog or pathway axis); used as specificity references only. **Dual?**: whether the evidence comes from simultaneous dual-gene perturbation. **Indep.**: evidence independent of DepMap CRISPR data. **Direct SL**: whether the evidence directly demonstrates synthetic lethality in the direction scored in this study (“Conditional” = genotype-conditional dependency; “Reciprocal” = direct evidence for the reverse direction only). **Inclusion**: Primary = primary external benchmark; Secondary = reported separately; Comparator = specificity reference. All twelve pairs are HGNC gene-group sequence paralogs except the two comparators; validated directions refer to the driver→paralog direction scored here. Key ref entries are short names corresponding to the numbered reference list in the main text.

3.4 External combinatorial-CRISPR gold standards (in4mer)

Esmaili Anvar et al. (2024) meta-analyzed five combinatorial CRISPR paralog screens (Dede 2020, Gonatopoulos-Pournatzis 2020, Parrish 2021, Thompson 2021, Ito 2021) and defined 13 candidate cross-study gold-standard paralog synthetic lethals (paralog score > 0.25 , called a hit in more than one study): CNOT7–CNOT8, PITPNA–PITPNB, TIA1–TIAL1, SAR1A–SAR1B, PTP4A1–PTP4A2, GSK3A–GSK3B, CSNK2A1–CSNK2A2, CSNK1D–CSNK1E, MAPK1–MAPK3, ARFGEF1–ARFGEF2, HDAC1–HDAC2, ASF1A–ASF1B, and SLC25A28–SLC25A37. We assessed whether these pairs could

serve as an external benchmark for DD. Only one of the 13 pairs (MAPK1–MAPK3) involves a gene in our 40-gene driver panel, and MAPK1 carries qualifying oncogenic hotspot mutations in zero DepMap 26Q1 cell lines (Table S6), so no in4mer gold-standard pair is evaluable in our driver-mutation-conditioned framework. This reflects a genuine division of scope rather than missing data: the in4mer gold standards measure background-independent paralog buffering under dual knockout, whereas DD quantifies dependency shifts conditioned on naturally occurring driver mutations. We therefore report the in4mer set as a conceptual cross-reference rather than a benchmark, and we note that only 8 of the 13 pairs replicated as hits in at least three of the four in4mer screens (Chou et al. 2025).

3.5 Table S4: 40-gene driver panel

Gene	Cancer Types	Source	Category
TP53	Pan-cancer	TCGA+COSMIC	TSG
KRAS	Lung, Colorectal, Pancreatic, Endometrial	TCGA+COSMIC	Oncogene
PIK3CA	Breast, Endometrial, Cervical, Colorectal	TCGA+COSMIC	Oncogene
PTEN	Endometrial, Ovarian, Breast, Glioma	TCGA+COSMIC	TSG
ARID1A	Endometrial, Ovarian, Colorectal	TCGA+COSMIC	TSG
BRCA1	Ovarian, Breast	TCGA+COSMIC	TSG
BRCA2	Ovarian, Breast	TCGA+COSMIC	TSG
BRAF	Melanoma, Colorectal, Thyroid	TCGA+COSMIC	Oncogene
NF1	Glioma, Breast, Lung	TCGA+COSMIC	TSG
RB1	SCLC, Breast, Bladder	TCGA+COSMIC	TSG
EGFR	Lung, Glioma, HNSCC	TCGA+COSMIC	Oncogene
ERBB2	Breast, Esophagogastric	TCGA+COSMIC	Oncogene
CDH1	Breast, Gastric	TCGA+COSMIC	TSG
CTNNB1	Endometrial, Hepatocellular	TCGA+COSMIC	Oncogene
STK11	Lung	TCGA+COSMIC	TSG
KEAP1	Lung	TCGA+COSMIC	TSG
EP300	Cervical, Colorectal	TCGA+COSMIC	TSG
FBXW7	Endometrial, Cervical, Colorectal	TCGA+COSMIC	TSG
PPP2R1A	Endometrial	TCGA+COSMIC	TSG
PIK3R1	Endometrial	TCGA	TSG
KMT2D	Endometrial, HNSCC	TCGA+COSMIC	TSG
CCNE1	Ovarian	TCGA+COSMIC	Oncogene
SMARCA4	Lung, Ovarian	TCGA+COSMIC	TSG
APC	Colorectal	TCGA+COSMIC	TSG
SMAD4	Colorectal, Pancreatic	TCGA+COSMIC	TSG
ATM	Pan-cancer	COSMIC	TSG
ATR	Pan-cancer	COSMIC	TSG
VHL	Renal	TCGA+COSMIC	TSG
SETD2	Renal	TCGA+COSMIC	TSG
BAP1	Mesothelioma, Renal	TCGA+COSMIC	TSG
PBRM1	Renal	TCGA+COSMIC	TSG
NOTCH1	HNSCC, Esophageal	TCGA+COSMIC	TSG
IDH1	Glioma	TCGA+COSMIC	Oncogene
SPOP	Prostate, Endometrial	TCGA+COSMIC	TSG
NFE2L2	Lung, Esophageal	TCGA+COSMIC	Oncogene
MET	Lung, Renal	TCGA+COSMIC	Oncogene
ALK	Lung	COSMIC	Oncogene
GATA3	Breast	TCGA	TSG
MAP3K1	Breast	TCGA	TSG
MYC	Pan-cancer	COSMIC	Oncogene

Table S4: Complete 40-gene driver panel (machine-readable mirror: `TableS4_DriverGenes.tsv`). Selection criteria: recurrently mutated in ≥ 1 TCGA pan-cancer study or COSMIC Cancer Gene Census, with ≥ 5 mutant cell lines in at least one DepMap lineage.

3.6 Table S5: All 21 paralog pairs ranked by dependency window score

Driver	Paralog	DD	DWS	Select.	Class	Type	DWS 95% CI	Select. 95% CI
NF1	RASA2	0.101	5.421	0.005	MOD	Seq	1.22–13.83	0.00–0.02
SMARCA4	SMARCA2	0.186	4.873	0.181	HS	Seq	3.28–7.78	0.07–0.32
RB1	RBL1	0.164	3.439	0.007	MOD	Seq	1.95–10.92	0.00–0.02
ARID1A	ARID1B	0.270	2.819	0.277	HS	Seq	1.76–5.89	0.19–0.37
EP300	CREBBP	0.279	1.815	0.108	MOD	Seq	1.13–4.55	0.00–0.21
KMT2D	KMT2C	0.064	1.771	0.056	MOD	Seq	0.94–5.00	0.02–0.11
ATR	ATM	0.070	1.656	−0.027	LS	Seq	0.31–4.79	−0.07–0.00
PIK3CA	PIK3CB	0.152	1.356	−0.065	LS	Seq	0.89–2.63	−0.11–0.01
BRAF	RAF1	0.245	1.174	−0.178	LS	Seq	0.85–1.70	−0.24–0.12
PPP2R1A	PPP2R1B	0.074	0.925	0.000	LS	Seq	0.09–2.22	0.00–0.00
TP53	TP63	0.035	0.813	0.037	MOD	Seq	0.42–3.04	0.01–0.07
FBXW7	FBXW2	0.025	0.720	0.000	LS	Seq	0.41–2.98	0.00–0.00
CDH1	CDH2	0.098	0.687	−0.086	LS	Seq	0.44–0.97	−0.10–0.07
KRAS	NRAS	0.083	0.420	−0.060	LS	Seq	0.26–0.83	−0.11–0.01
STK11	SIK1	0.017	0.392	−0.011	LS	Part	0.03–1.27	−0.02–0.01
KRAS	HRAS	0.043	0.206	−0.040	LS	Seq	0.11–0.43	−0.08–0.01
BRCA2	BRCA1	0.111	0.193	−0.047	PE	Func	0.13–0.45	−0.23–0.14
PTEN	TNS2	0.042	0.174	−0.006	LS	Seq	0.11–0.39	−0.07–0.07
PIK3R1	CRKL	0.152	0.167	−0.161	PE	Seq	0.08–0.71	−0.36–0.05
BRCA1	BRCA2	0.079	0.144	−0.080	PE	Func	0.05–0.40	−0.30–0.14
MAP2K1	MAP2K2	0.001	0.003	−0.034	LS	Seq	0.01–0.51	−0.04–0.02

Table S5: All 21 analyzed paralog pairs ranked by mean dependency window score (DWS) on the ≥ 5 -mutant frame. |DD|: mean absolute Delta Dependency across cancer contexts. Select.: mean selectivity (fraction mutant-essential minus fraction wild-type-essential). Class: HS=HIGH_SELECTIVITY, MOD=MODERATE, LS=LOW_SELECTIVITY, PE=PAN_ESSENTIAL. Type: Seq=sequence paralog, Part=partial homology, Func=functional analog. DWS 95% CI and Select. 95% CI: bootstrap 95% percentile intervals from 1,000 stratified resamples of cell lines within each cancer context (seed 42; `dws_robustness.py`); the DWS and Select. columns report the observed point estimates, not bootstrap means. Three pairs evaluable under the previous frame (AKT1→AKT2, CCNE1→CCNE2, CDK4→CDK6) no longer meet the ≥ 5 -mutant threshold in any context and are not listed.

3.7 Table S6: Gene-class-specific driver-mutation rules and variant classification

Gene	Class	Rule	Old	New	% ret.
ALK	ONC	Hotspot	118	12	10.2
APC	TSG	LikelyLoF	217	108	49.8
ARID1A	TSG	LikelyLoF	240	173	72.1
ATM	TSG	LikelyLoF	165	59	35.8
ATR	TSG	LikelyLoF	125	37	29.6
BRAF	ONC	Hotspot	212	162	76.4
BRCA1	TSG	LikelyLoF	101	34	33.7
BRCA2	TSG	LikelyLoF	159	68	42.8
CCNE1	ONC	Hotspot	22	0	0.0
CDH1	TSG	LikelyLoF	87	34	39.1
CDK4	ONC	Hotspot	27	8	29.6
CTNNB1	ONC	Hotspot	85	45	52.9
EGFR	ONC	Hotspot	110	19	17.3
EP300	TSG	LikelyLoF	206	107	51.9
ERBB2	ONC	Hotspot	70	20	28.6
FBXW7	TSG	LikelyLoF	116	81	69.8
GATA3	TSG	LikelyLoF	52	16	30.8
KEAP1	TSG	LikelyLoF	99	31	31.3
KMT2D	TSG	LikelyLoF	356	185	52.0
KRAS	ONC	Hotspot	254	233	91.7
MAP2K1	ONC	Hotspot	36	14	38.9
MAP3K1	TSG	LikelyLoF	74	14	18.9
MAPK1	ONC	Hotspot	13	0	0.0
MET	ONC	Hotspot	66	1	1.5
NF1	TSG	LikelyLoF	215	122	56.7
PIK3CA	ONC	Hotspot	182	153	84.1
PIK3R1	TSG	LikelyLoF	79	46	58.2
PPP2R1A	TSG	LikelyLoF	53	13	24.5
PTEN	TSG	LikelyLoF	199	176	88.4
RB1	TSG	LikelyLoF	185	150	81.1
SMAD4	TSG	LikelyLoF	119	80	67.2
STK11	TSG	LikelyLoF	97	66	68.0
TP53	TSG	LikelyLoF	1171	1140	97.4

Table S6: Driver-mutation definitions by gene class (DepMap 26Q1, default-entry profiles). TSG: mutant requires **LikelyLoF**; ONC: mutant requires **Hotspot**. Old: cell lines carrying any non-silent variant (pre-revision permissive definition); New: cell lines qualifying under the class-specific rule; % ret.: $100 \times \text{New}/\text{Old}$. The top VariantInfo terms per gene are at `output/driver_mutation_rules.csv`. CCNE1 and MAPK1 are amplification-driven oncogenes with no qualifying hotspot mutations.

3.8 Table S8: TCGA BRCA overall-survival associations for the 32 paralog genes

Gene	<i>n</i>	Univariable			Multivariable (age + AJCC stage)			PH <i>p</i>
		HR (95% CI)	<i>p</i>	<i>q</i>	HR (95% CI)	<i>p</i>	<i>q</i>	
<i>PIK3CA</i>	1069	1.22 (1.03–1.46)	0.024	0.383	1.34 (1.12–1.59)	0.001	0.035	0.239
<i>RBL1</i>	1069	1.13 (0.96–1.34)	0.135	0.513	1.30 (1.09–1.56)	0.003	0.040	0.359
<i>ARID1B</i>	1069	1.30 (1.09–1.55)	0.004	0.129	1.31 (1.09–1.57)	0.004	0.040	0.271
<i>BRCA2</i>	1069	1.10 (0.93–1.29)	0.257	0.513	1.27 (1.08–1.50)	0.005	0.040	0.695
<i>STK11</i>	1069	0.90 (0.77–1.05)	0.167	0.513	0.85 (0.72–0.99)	0.038	0.210	0.151
<i>RAF1</i>	1069	1.10 (0.94–1.28)	0.253	0.513	1.19 (1.01–1.40)	0.039	0.210	0.924
<i>PIK3CB</i>	1069	1.12 (0.96–1.30)	0.144	0.513	1.16 (0.99–1.36)	0.071	0.323	0.793
<i>RASA2</i>	1069	1.03 (0.86–1.24)	0.741	0.891	1.19 (0.97–1.44)	0.089	0.328	0.047
<i>BRCA1</i>	1069	1.16 (0.99–1.36)	0.066	0.513	1.15 (0.97–1.36)	0.100	0.328	0.353
<i>KRAS</i>	1069	1.15 (0.97–1.35)	0.113	0.513	1.16 (0.97–1.38)	0.103	0.328	0.750
<i>SMARCA4</i>	1069	1.05 (0.90–1.24)	0.531	0.823	1.13 (0.97–1.33)	0.122	0.349	0.061
<i>TNS2</i>	1069	0.96 (0.82–1.11)	0.562	0.823	0.89 (0.77–1.04)	0.131	0.349	0.156
<i>PPP2R1B</i>	1069	1.01 (0.85–1.21)	0.876	0.916	1.12 (0.93–1.34)	0.237	0.583	0.570
<i>ATR</i>	1069	0.99 (0.84–1.17)	0.892	0.916	1.09 (0.92–1.29)	0.321	0.620	0.633
<i>FBXW2</i>	1069	1.10 (0.93–1.30)	0.254	0.513	1.09 (0.92–1.30)	0.331	0.620	0.011
<i>FBXW7</i>	1069	1.01 (0.86–1.19)	0.908	0.916	1.08 (0.92–1.27)	0.337	0.620	0.066
<i>HRAS</i>	1069	0.90 (0.76–1.06)	0.200	0.513	0.92 (0.78–1.09)	0.338	0.620	0.040
<i>PPP2R1A</i>	1069	1.09 (0.94–1.27)	0.254	0.513	1.08 (0.92–1.27)	0.360	0.620	0.483
<i>ATM</i>	1069	0.99 (0.84–1.17)	0.916	0.916	1.08 (0.91–1.28)	0.368	0.620	0.071
<i>BRAF</i>	1069	0.90 (0.76–1.06)	0.199	0.513	0.93 (0.78–1.10)	0.404	0.632	0.391
<i>CREBBP</i>	1069	0.96 (0.83–1.11)	0.566	0.823	0.94 (0.81–1.09)	0.415	0.632	5.5×10^{-4}
<i>RB1</i>	1069	1.10 (0.93–1.30)	0.276	0.520	1.06 (0.89–1.26)	0.500	0.727	0.365
<i>KMT2D</i>	1069	1.05 (0.88–1.25)	0.592	0.823	1.05 (0.88–1.25)	0.576	0.772	0.135
<i>PTEN</i>	1069	0.97 (0.83–1.14)	0.730	0.891	1.05 (0.88–1.25)	0.586	0.772	0.024
<i>SMARCA2</i>	1069	0.90 (0.76–1.06)	0.205	0.513	0.96 (0.81–1.13)	0.603	0.772	0.174
<i>PIK3R1</i>	1069	0.88 (0.75–1.05)	0.149	0.513	0.96 (0.81–1.14)	0.671	0.809	0.282
<i>ARID1A</i>	1069	1.03 (0.88–1.20)	0.752	0.891	1.03 (0.88–1.22)	0.701	0.809	0.002
<i>NF1</i>	1069	1.07 (0.90–1.28)	0.457	0.812	1.03 (0.86–1.24)	0.732	0.809	0.014
<i>SIK1</i>	1069	0.87 (0.74–1.02)	0.089	0.513	0.97 (0.83–1.14)	0.740	0.809	0.604
<i>CRKL</i>	1069	0.95 (0.79–1.13)	0.549	0.823	1.03 (0.86–1.23)	0.759	0.809	0.048
<i>KMT2C</i>	1069	1.01 (0.85–1.20)	0.873	0.916	1.02 (0.85–1.22)	0.833	0.860	0.022
<i>EP300</i>	1069	1.03 (0.86–1.23)	0.751	0.891	1.01 (0.85–1.21)	0.881	0.881	0.007

Table S8: Cox proportional-hazards associations between paralog gene expression (continuous, $\log_2$ -transformed, z-scored; HR per +1 SD) and overall survival in the TCGA PanCan Atlas BRCA cohort ($n = 1,069$, 151 events). Multivariable models adjust for age and AJCC stage (ordinal I–IV; complete cases $n = 1,050$, 143 events). *q*: Benjamini–Hochberg FDR across the 32-gene family within each model class; four genes pass $\text{FDR} < 0.05$ in the multivariable family (*PIK3CA*, *RBL1*, *ARID1B*, *BRCA2*) and none in the univariable family. PH *p*: Schoenfeld-type residual–time correlation test of the proportional-hazards assumption (approximation without variance scaling); genes with PH $p < 0.05$ should be interpreted with caution. Sorted by multivariable *p*. Source: `output/tcga_survival_v2.json` (`tcga_survival_v2.py`).

3.9 Table S9: Top paralog-SL candidates ranked by composite prioritization score

Table S9: Top paralog-SL candidates ranked by the exploratory composite prioritization score (moved from the main text when the structural analysis was demoted to exploratory descriptor status). DWS: mean dependency window score across up to 5 cancer contexts on the ≥ 5 -mutant frame. Class: HS = HIGH_SELECTIVITY, MOD = MODERATE, LS = LOW_SELECTIVITY, PE = PAN_ESSENTIAL. Struct: feature-space structural similarity (heuristic, not a 3D alignment metric; Methods). Target.: composite prioritization score (max-normalized DWS 0.40, rescaled selectivity 0.30, targetability 0.30; weights fixed a priori in `alphafold_analysis.py`). Known SL pairs marked with $\star$. *NF1* $\rightarrow$ *RASA2*'s top composite score is driven by a near-zero pan-essential denominator (0.1% of cell lines) that inflates its DWS; with selectivity ≈ 0 it is not a selective candidate, and *ARID1A* $\rightarrow$ *ARID1B* remains the leading selective candidate. These descriptors were not calibrated against drug-development data; all candidates are computationally nominated and require experimental validation.

Driver	Paralog	DWS	Class	Known	Struct	Target.
NF1	RASA2	5.42	MOD		0.658	0.695
ARID1A	ARID1B	2.82	HS	$\star$	0.952	0.631
EP300	CREBBP	1.82	MOD	$\star$	0.976	0.535
KMT2D	KMT2C	1.77	MOD		0.829	0.477
PIK3CA	PIK3CB	1.36	LS	$\star$	0.847	0.471
PPP2R1A	PPP2R1B	0.93	LS	$\star$	0.921	0.452
TP53	TP63	0.81	MOD		0.803	0.393
KRAS	HRAS	0.21	LS		0.885	0.392
FBXW7	FBXW2	0.72	LS	$\star$	0.666	0.358
STK11	SIK1	0.39	LS	$\star$	0.710	0.334

3.10 Table S10: Sensitivity of dependency-window score rankings and selectivity classification

Denominator variant	Spearman ρ vs. base	Top-5 overlap
Base: $\max(\mu , f, 0.01)$ (production)	1.000	5
$ \mu $ only	0.990	4
f (pan-essential fraction) only	0.893	4
$\text{mean}(\mu , f)$	0.988	4
Floor 0.001	1.000	5
Floor 0.05	0.974	5
HIGH_SELECTIVITY selectivity threshold	HS pairs	Flips vs. 0.15
Selectivity > 0.10	3	1
Selectivity > 0.15 (base)	2	0
Selectivity > 0.20	1	1

Table S10: Sensitivity of DWS-based rankings to the denominator definition (upper block) and of the HIGH_SELECTIVITY classification to the selectivity threshold (lower block), for the 21 evaluated paralog pairs. Spearman ρ : rank correlation of the DWS ranking against the production definition; top-5 overlap: number of shared pairs in the top five of both rankings. At threshold 0.10, EP300 $\rightarrow$ CREBBP additionally enters HIGH_SELECTIVITY; at 0.20, SMARCA4 $\rightarrow$ SMARCA2 drops out; ARID1A $\rightarrow$ ARID1B remains HIGH_SELECTIVITY at all three thresholds. Source: `output/dws_robustness.json` (`dws_robustness.py`; 1,000 stratified bootstrap resamples, seed 42).